Counterfactual Prices, Inflation, and Adoption: A Voluntary Price Cap in the Italian Retail Fuel Market

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Abstract

When international oil prices were rising steeply, Italy's largest fuel retailer voluntarily capped its pump prices from 28 September 2026 at a nominal 1.99 €/l for petrol and 2.19 €/l for diesel, and most of the market followed within weeks. We reconstruct the effective daily price of every roadside Italian fuel station from the official price-communication registry, strip the statutory tax calendar, and compare the observed price path with a no-cap counterfactual built from pre-cap oil pass-through estimated over January–September 2026. The counterfactual projects October pump prices 17.4 c/l (petrol) and 21.1 c/l (diesel) above realised prices on average, a range that survives credible specification and forecast-validation checks. Evaluated at the statistical office's reference days, the cap turned fuel inflation from +5.3 and +10.7% (petrol and diesel, no cap) into −2.3 and +1.6%, an estimated contribution of about −0.4 percentage points to October headline consumer-price inflation. The station-level panel explains how the aggregate effect arose: the leader cut most, branded rivals matched within days, prices bunched one tenth of a cent below the ceiling, and the 28 October diesel excise cut was passed through only gradually at the ceiling. A time-of-day reconstruction correction of up to 5 c/l matters for reference-day price measurement.

1 Introduction

Between January and October 2026 the price of Brent crude in euro doubled, from roughly 53 to 108 € per barrel. In the middle of this surge, on 25 September 2026, Eni—the company operating Italy's largest fuel-retail network—announced that from the following Monday, 28 September, it would cap the self-service prices at its stations at 1.99 € per litre of petrol and 2.19 € per litre of diesel, roughly 17 cents below the prevailing market level, initially for thirty days with a possible extension to the end of the year. Rival networks matched within days and the commitment on the cap level was maintained, by the company's communications, through the end of October, the window we study. A voluntary, firm-initiated price ceiling during an inflationary oil shock is an unusual event, and it poses a well-defined question: how much lower were Italian pump prices than they would otherwise have been, and what did that difference contribute to measured consumer-price inflation?

Answering that question requires measuring three things. First, the *observed* national price path, which we construct from the complete registry of Italian station price communications—12.4 million station–fuel–day observations covering 21,766 stations over 2026—reconstructing for every station and day the price actually in force rather than the price recorded by the daily extract. Second, a *conditional no-cap path*: an econometric model of how national net-of-tax pump prices passed through the Brent price before the cap, projected over the cap window on the actual oil path with the actual statutory taxes reapplied. The counterfactual is conditional on the maintained assumption that pre-cap pass-through would have continued absent the cap; it is not an experimentally identified quantity, and we discipline it with forecast validation and specification bounds rather than claim identification. Third, an *inflation-accounting implication*: the difference between the observed and no-cap paths, evaluated on the 1st, 11th and 21st of September and October—the reference days on which Italy's statistical office (ISTAT) collects fuel prices for the consumer-price index—and aggregated with explicit, documented assumptions.

Three findings emerge. The observed effect is large: averaging over the 34 post-cap days, the no-cap model projects October prices 17.4 c/l above realised prices for petrol and 21.1 c/l for diesel; on 31 October the gaps are 22.2 and 20.4 c/l. Because prices are collected on reference days, this price difference maps into measured inflation: month-on-month fuel prices fell 2.3% (petrol) and rose 1.6% (diesel) with the cap, against a no-cap counterfactual of +5.3 and +10.7%, an estimated contribution of −0.38 percentage points to October headline inflation (specification range −0.38 to −0.45). The propagation is leader-led: on the first day 21% of stations priced below the ceiling, half within four days, and 83–84% at some point by 31 October; the initiator's network accounted for a disproportionate share of the total price decline, unbranded independents the smallest. And the ceiling reshaped the price distribution: a large mass of stations priced one tenth of a cent below it, the ninetieth–tenth percentile spread of within-province prices rose by about seventy per cent, and when a diesel excise cut shifted the ceiling down on 28 October the network passed the tax change through only gradually—a small natural experiment in ceiling-constrained tax incidence.

Our contribution is primarily one of measurement and interpretation of an aggregate policy episode, and the paper speaks to four literatures. To the large body of work on oil-to-retail pass-through and asymmetric pricing (Atil et al., 2013; Bettendorf et al., 2003; Borenstein & Shepard, 2002; Tappata, 2009), we add a case in which pass-through was suspended not by regulation of margins but by a firm's own price commitment. To work on price ceilings in retail fuel markets—statutory ceilings, their focal-point effects and their interaction with market structure (Carranza et al., 2015; Contín-Pilart et al., 1999; Gatsios et al., 2026; Quiguanas et al., 2025; Sen et al., 2011; Zhang et al., 2020)—we add a voluntary, leader-initiated ceiling with near-complete market matching, observed at daily frequency. To the literature on price leadership and price matching (Amir & Stepanova, 2008; Atkinson et al., 2009; Kim et al., 2019; Pellegrino, 2025), we provide a clean episode in which the follower response can be observed station by station. Finally, to work on price-index measurement with administrative data (Pennerstorfer et al., 2020; Rossi & Chintagunta, 2015), we show that reconstructing prices by communication time rather than extract time corrects reference-day averages by up to 5 c/l on high-churn days, a distinction that matters for any statistics built on the Italian registry.

The paper proceeds as follows. Section 2 describes the institutional setting, including the 2026 excise calendar that any analysis of Italian fuel prices must respect. Section 3 documents the data and the effective-price reconstruction. Section 4 presents the observed market response. Section 5 builds and validates the no-cap counterfactual; Section 6 translates it into an estimated inflation contribution. Section 7 examines how the response propagated across stations, brands and geography. Section 8 concludes.

2 Institutional setting

Italian road fuel is sold through roughly 21,000 reporting roadside (*stradale*) stations under a mix of brand banners—Eni's network, the other large brands (Q8, Api-Ip, Eni's former competitors Esso and Tamoil among them)—and unbranded "white pump" independents, alongside some 2,000 motorway stations whose prices are regulated at higher levels and which we exclude throughout. Since 2005 every price change must be communicated to the Ministry of Enterprises (MIMIT), which publishes the full registry daily as open data; the price in force at any moment is therefore a matter of public record. Italy's consumer-price statistics draw fuel prices from this same registry, observing them on three reference days per month—the 1st, the 11th and the 21st.

Two features of the 2026 tax calendar are essential context. First, excise duty was itself on a roller-coaster: a 20 c/l cut for both fuels took effect on 19 March, and the discount was phased back in through a sequence of steps between 2 May and 4 July, with a further diesel-only discount from 28 July that was itself withdrawn in steps on 18 and 26 September, days before the cap. Second, the cap was explicitly framed by the announcing company as linked to the excise relief

in force: when the last 5 c/l of diesel discount was withdrawn on 6 October, the gross ceiling mechanically translated upward with it, and when a 4 c/l diesel excise cut took effect on 28 October the ceiling translated downward. Any compliance or price-gap measure must therefore use a date-specific ceiling, which we construct below. Petrol excise was unchanged after 4 July; in particular there is no petrol tax change in the outcome window. Value-added tax is 22% throughout, so a one-cent excise change moves the gross pump price by 1.22 cents.

The company's announcement (25 September, effective 28 September) stated nominal ceilings of 1.99 and 2.19 €/l and an initial duration of thirty days, with extension to year-end possible depending on market conditions. Prices in the registry are quoted to three decimals. Our operational compliance cutoffs are the rounded-up gross levels of 2.000 €/l (petrol) and 2.200 €/l (diesel): a price of 1.999 or 2.199 complies. We apply the cutoff strictly—a price exactly at 2.000 or 2.200 does not comply—which is empirically nearly inconsequential: exact-boundary observations never exceed 0.06% of station-days (table 8). The diesel ceiling steps to 2.261 on 6 October (2.200 plus the withdrawn 5 c/l excise grossed up by VAT) and to 2.212 on 28 October (2.261 minus the 4 c/l cut grossed up). Throughout, the analysis window ends on 31 October, by which date the hard commitment on the cap level had run its course; communications about pricing beyond that date were less specific, and we do not analyse them.

3 Data and measurement

3.1 Sources

We use four sources. The MIMIT price registry, as daily extracts covering 1 January to 5 November 2026: 307 extract days (16–17 October were never published, the only gap) and 12.4 million rows for self-service petrol and diesel. Weekly station snapshots provide each station's banner, operator, type and location. The statutory excise calendar, obtained from the historical tax registry back to 2005, gives every application date and rate. Brent in euro is the FRED Brent spot price converted at the USD/euro reference rate, forward-filled to calendar days. Table 1 summarises the resulting samples.

3.2 From communications to effective prices

The registry extract published on a given day contains, for every station and fuel, the last communicated price—and each row carries the timestamp at which that price was communicated. The extract therefore mixes prices set at very different times, and on high-churn days it can misrepresent the price in force on any particular day. The problem is sharpest exactly where it matters for us: the cap was implemented through a wave of communications in the early morning of 28 September, so the extract published that day still shows, for most stations, the pre-cap price, while the extract of 1 October straddles the following repricing wave.

We therefore reconstruct the *effective* price of every station–fuel pair on every day: the price of the last communication dated on or before that day. A station enters the panel on its first in-window communication and leaves when it stops being observed in extracts, so stale prices are never carried forward indefinitely. The practical stakes are large. On 1 October 2026 the mean extract price exceeded the mean effective price by 4.2 c/l (petrol) and 4.9 c/l (diesel), because the morning extract cut off mid-wave; a reference-day statistic built on extract-time prices would overstate that day's price by that amount. All our price series are effective-price series; we return to the measurement implications in Section 6. MIMIT published no extracts for 16–17 October 2026; because prices are assigned by communication date, the effective price of those two days is recovered from the communications carried by the surrounding extracts, so the reconstructed path covers every calendar day of the window.[1]

[1]The only information possibly lost to the missing downloads is a price change communicated and fully superseded within the two-day gap; such same-week double repricing is rare, and the reconstructed 16–17 October

3.3 From pump prices to net-of-tax prices

For the counterfactual we work with the net-of-tax price, defined as

$$p_{st}^{net} = \frac{p_{st}}{1+v} - \tau_t,$$

where p_{st} is the effective gross price at station s on day t, $v = 0.22$ is the VAT rate and τ_t the statutory excise for that fuel on day t. Netting the tax removes every 2026 excise step—the spring cut, its phased reinstatement, the pre-cap diesel steps and the two post-cap diesel steps—from both sides of the pass-through relationship by construction, so no tax dummy belongs in the model. The net-of-tax price is a price received along the supply chain, not a retail margin: the wedge between it and the wholesale cost is shared between refining, distribution and retailing, and we make no claim about that split. The national series we analyse are equal-weighted means of effective prices across roadside self-service stations, about 19,700 per fuel per day, with a balanced-panel variant (stations present on at least 95% of days) and an ISTAT-like regularity-filtered variant (stations that communicated at least forty distinct price changes over July–September, 57.5% of units) used as composition checks.

3.4 Compliance and adoption

A station–fuel pair is defined as pricing below the ceiling on day t if its effective gross price is strictly below the date-specific ceiling of Section 2. Three distinct objects should not be confused: the share of observations below the ceiling on a given day; the share of units that have *ever* priced below it since 28 September (our adoption measure); and formal enrolment in any company's initiative, which we do not observe and do not require—pricing below the ceiling is a behaviour, not a membership. The estimation panel for adoption comprises 39,425 station–fuel units observed from 28 September to 31 October (4.8 million station-days), of which 32,974 adopt at some point.

4 Observed market response

Figure 1 shows the national price paths in their 2026 context. The spring excise cut pulled both fuels down by roughly 20 c/l in March; the phased reinstatement lifted them through the summer while oil was already rising; and at the moment of the cap in late September diesel stood at 2.374 €/l and petrol at 2.158 €/l, both close to their yearly highs. Over the cap window, petrol fell to 2.009 by 31 October (−14.9 c/l) and diesel to 2.239 (−13.5 c/l). The diesel path is the harder to read, because two excise steps fall inside the window: the 6 October increase (+6.1 c/l gross) and the 28 October cut (−4.9 c/l gross). In net-of-tax terms—which strips both—petrol fell 12.2 and diesel 12.1 c/l over the window, against a Brent price that rose further.

Adoption was fast and broad. On the first day, 20.9% of petrol and 21.4% of diesel observations were already priced below the ceiling; by the fourth day (2 October) roughly half were; by 31 October 84.4% of petrol units and 82.8% of diesel units had ever priced below their date-specific ceiling, with a median first-compliance day of 4. Two compositional notes are essential. First, final-day and ever-shares have different denominators: on 31 October itself, 81.8% of petrol and 50.2% of diesel *observations* were below the ceiling, the latter lower because the corrected 2.212 ceiling of the last three days is harder to meet, as Section 7 shows. Second, no measurable anticipation preceded implementation: in the event-study of Section 7 the pre-cap coefficients for 21–27 September are between −0.2 and −0.4 percentage points, an order of magnitude smaller than the 19-point jump on day zero, and the announcement-to-implementation gap of 25–27 September shows no break in the aggregate price path.

means move smoothly between their neighbours. The loss of information is marginal and does not affect any result in the paper.

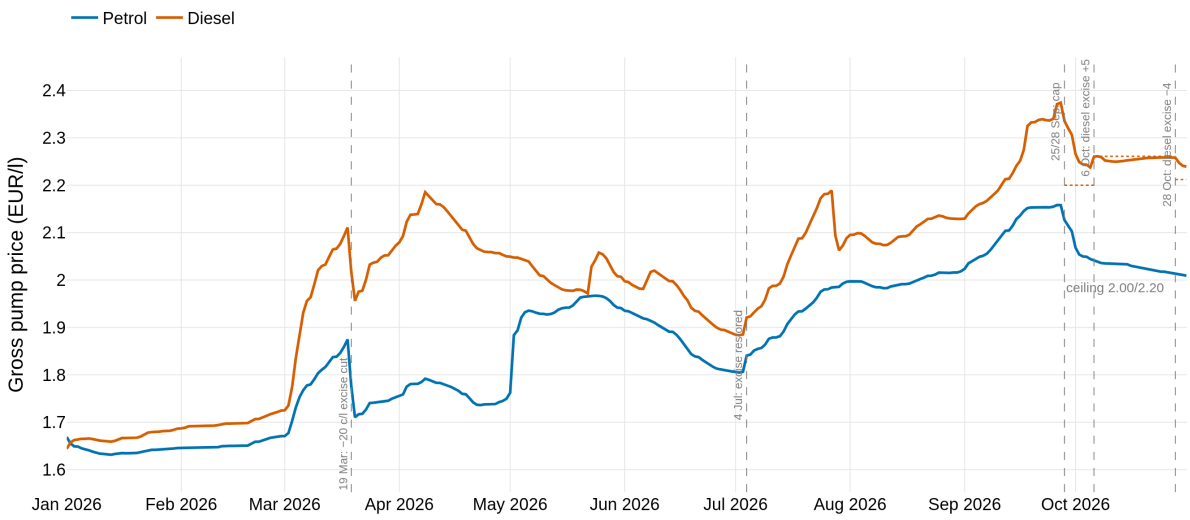


Figure 1: National mean gross pump prices (self-service, roadside), January to October 2026. Dashed vertical lines mark the 19 March excise cut, the 4 July restoration, the cap (announcement 25, implementation 28 September) and the two post-cap diesel excise steps. The dotted stepped line is the operational diesel ceiling (2.200 from 28 September, 2.261 from 6 October, 2.212 from 28 October); the petrol ceiling is 2.000 throughout. The 16–17 October extracts were never published; those days are recovered from the surrounding communications (see section 3).

Brand groups ordered the response exactly as a leadership account would predict. Eni's network was at the ceiling almost immediately and stayed there; Q8 and Api-Ip matched within a week; Esso and Tamoil followed slightly later; and unbranded independents converged only partially. A simple descriptive decomposition of the 27 September–31 October mean price decline (fig. 4a) attributes 27% of the diesel decline to Eni's network—which accounts for about 19% of observations—and only 4% to white pumps, whose prices fell least. The 6 October diesel episode separates mechanics from behaviour cleanly: of the 48-point jump in the share of diesel observations inside the newly compliant band [2.200, 2.261), 43 points are stations that actively re-communicated a price that day; only 5 points are stations whose pre-existing price merely found itself inside the widened band. The compliance jump was overwhelmingly repricing, not reclassification.

5 A no-cap price counterfactual

5.1 Strategy

The counterfactual question is what national prices would have been over 28 September–31 October had the cap not happened, given the oil path that actually occurred. Our answer conditions on the pre-cap relationship between net-of-tax pump prices and the Brent price in euro, estimated on the extended window from 2 January to 27 September 2026—261 daily observations spanning the spring excise cut, its reinstatement and the pre-cap oil surge, all of which are absorbed by the net-of-tax construction. The maintained assumption is that this pass-through relationship would have continued; everything that follows is an attempt to make that assumption transparent, to test where it can be tested, and to bound the conclusions where it cannot.

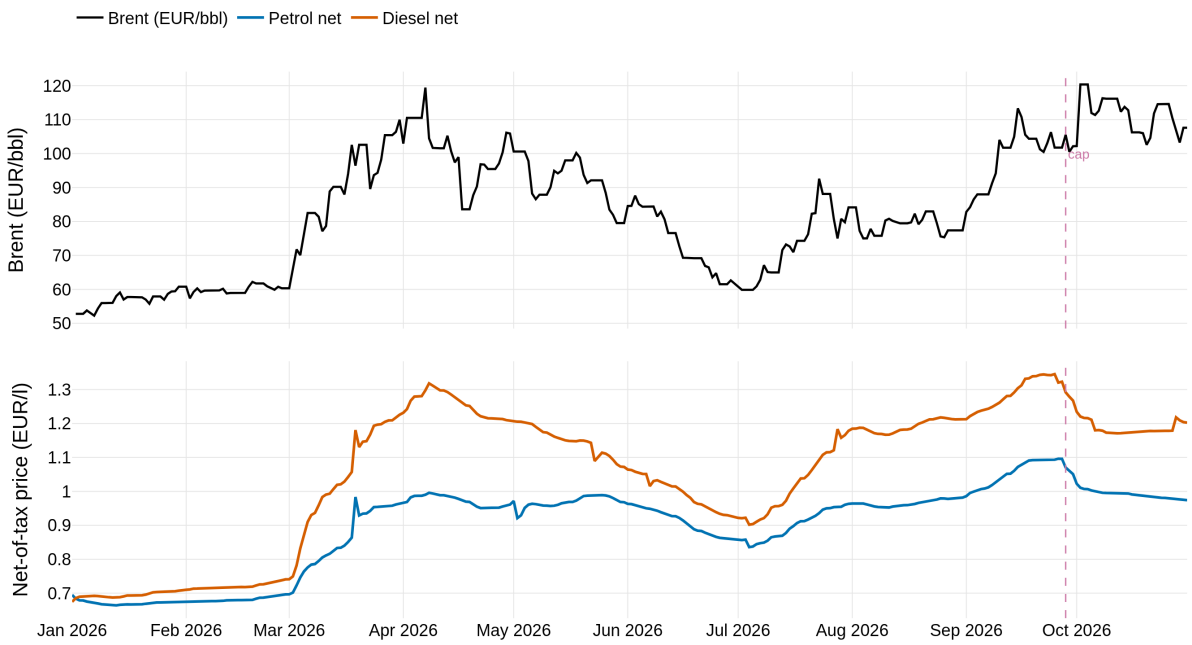


Figure 2: Brent in euro (top) and national mean net-of-tax pump prices (bottom), January to October 2026. The estimation window for the pass-through model ends on 27 September (dashed line).

5.2 Model and selection

For each fuel we estimate an autoregressive distributed-lag (ARDL) model in logs,

$$\ln p_t^{net} = \alpha + \sum_{i=1}^{q} \phi_i \ln p_{t-i}^{net} + \sum_{j=0}^{q} \theta_j \ln b_{t-j} + \varepsilon_t,$$

where b_t is Brent in euro and lag orders are selected by the Bayesian information criterion over an exhaustive grid: own lags $p \in [0, 21]$ and oil lags $q \in [7, 21]$, with the floor of seven oil lags imposing at least one week of Brent dynamics in the polynomial. The floor reflects a substantive prior—retail prices adjust to oil over days, not overnight—and we verify below that the results are insensitive to it. On the extended window, BIC selects $(p, q) = (2, 8)$ for petrol and $(1, 8)$ for diesel (table 2).

5.3 Validation

Three exercises discipline the model. First, a placebo backtest: refit on data through 13 September and project recursively over 14–27 September, still pre-cap. The model attains a root-mean-square error of 0.45 c/l (petrol) and 0.85 c/l (diesel) on the net price, against 3.69 and 5.07 c/l for a no-change random walk, with biases near zero (table 2). Second, expanding-origin backtests: five origins from 31 May to 13 September, projecting 14 and 28 days ahead, all evaluation windows strictly pre-cap. The oil model beats the random walk in eleven of sixteen origin–horizon cells; the losses concentrate at 14-day horizons from the mid-summer origins, when prices drifted with little oil news. We report these losses rather than select around them; the origins closest to the counterfactual period—late August and September—favour the oil model decisively. Third, block-exclusion tests with heteroskedasticity-and-autocorrelation-robust covariance reject the exclusion of the Brent block for both fuels ($p < 10^{-10}$). A pure autoregression without oil terms is reported as a mechanical persistence benchmark: on the extended window it forecasts *worse* than the oil models for both fuels (backtest 3.00 and 4.31 c/l), which was not true on the shorter summer-only window and is one reason we prefer the extended estimation sample.

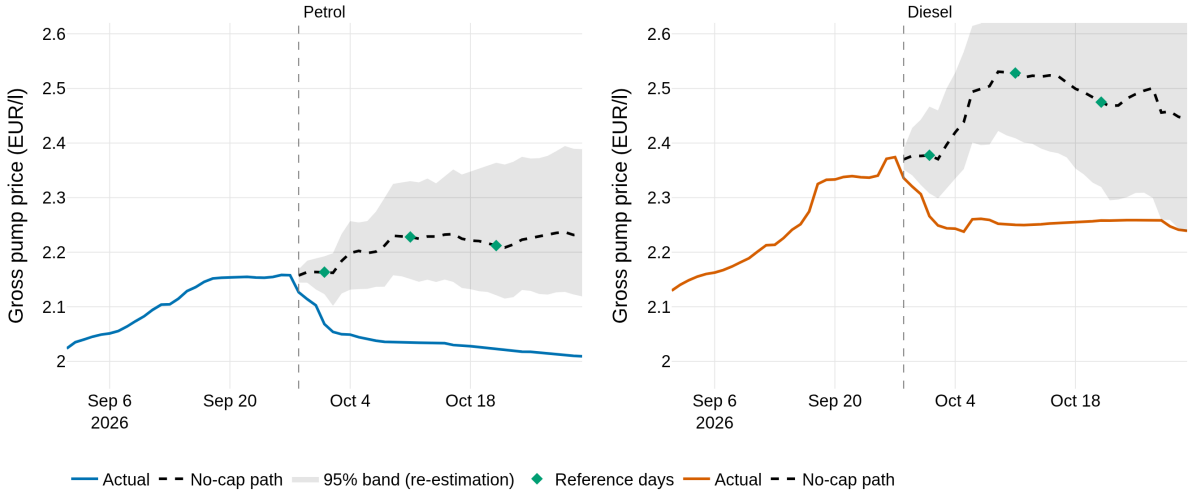


Figure 3: Actual (solid) and conditional no-cap (dashed) gross pump prices, September–October 2026, with 95% bands from a re-estimation bootstrap (400 draws: resampled residuals, re-estimated coefficients, re-projected paths). Diamonds mark the 1st/11th/21st reference days. The vertical line is the 28 September implementation.

One diagnostic deserves honest statement. Both selected specifications carry near-unit-root persistence (the sum of own-lag coefficients is 0.987 for petrol and 0.995 for diesel), so the implied long-run elasticity of the diesel main specification—0.09—is a small-denominator artifact and not a structural estimate. Across the admissible oil-lag floors, long-run elasticities range from 0.72 to 0.77 for petrol and 0.81 to 0.86 for diesel; Pesaran–Shin–Smith bounds tests (appendix B) cannot reject the absence of a levels relationship for either fuel over this window. The credibility of the counterfactual rests on its validated forecasting behaviour, not on a point estimate of long-run pass-through.

5.4 The no-cap path

Figure 3 presents the projection. To obtain a no-cap *gross* price we re-apply the statutory tax calendar to the projected net path—the same excise on the same days as in the observed series—so the 6 and 28 October diesel tax steps affect both paths identically. The counterfactual petrol price starts where the market started, about 3 c/l above the realised price on 28 September, and climbs with Brent to 2.233 €/l by 31 October, 22.4 c/l above the actual price; the diesel counterfactual reaches 2.444, a gap of 20.5 c/l. Averaged over the window, the no-cap path lies 17.4 c/l (petrol) and 21.1 c/l (diesel) above the realised one. On the inflation reference days the gaps are −9.5, −19.3 and −18.9 c/l (petrol; 1/11/21 October) and −11.2, −27.8 and −21.7 c/l (diesel).

Uncertainty is reported in layers, each covering a distinct component. Residual-shock bootstrap bands (2,000 draws) reflect innovation uncertainty conditional on the estimated coefficients; moving-block bootstrap bands (block length seven) add serial dependence in the shocks; a re-estimation bootstrap (400 draws of resampled residuals, re-estimated coefficients, re-projected paths) adds coefficient uncertainty and is the widest—by 31 October it spans the petrol gap from −11 to −38 c/l and the diesel gap from −0.2 to −44 c/l. Model uncertainty is deliberately kept outside the bands and shown instead as a specification range: across oil-lag floors, fit windows (estimating from July instead of January moves the diesel gap from −21.1 to −24.5 c/l), balanced and regularity-filtered samples, and Brent paths shifted by ±5%, October mean gaps lie between 15 and 19 c/l for petrol and 21 and 24 c/l for diesel, with the distributed-lag-only diesel variant an outlier at −26.1 (table 6). Under every admissible oil-containing specification the no-cap price is materially above the realised price throughout the window; the direction of

the conclusion does not depend on the specification, only its size does.

6 Inflation implications

ISTAT measures fuel prices for the consumer-price index (NIC) on the 1st, 11th and 21st of each month, drawing on the same MIMIT registry. We compute month-on-month September-to-October changes of the mean reference day price for each fuel, once as observed and once along the no-cap path, and aggregate with explicit assumptions: the ECOICOP 07.2.2 class "carburanti e lubrificanti" carries a 2026 basket weight of 4.5219% (verified against the published weighting structure; the class includes lubricants, a small residual we do not separately model), and we split it evenly between petrol and diesel, reflecting their approximately equal contributions to the class. Station observations enter with equal weights; a regularity-filtered variant mimicking the office's preference for stations with a consistent communication history serves as the composition check. These are documented approximations to an aggregation procedure the statistical office does not publish in full; we claim an *estimated contribution under stated assumptions*, not a replication of the official calculation.

Table 1: Data and sample definitions

Item	Value	Notes
Price communications (rows)	12.4 million	1 Jan–5 Nov 2026
Stations reporting prices	21 766	unique MIMIT plant IDs
Extract days	307	none 16–17 Oct
Station–fuel units (micro panel)	39 425	28 Sep–31 Oct
Estimation observations	261 per fuel	2 Jan–27 Sep, after lags
Reference days	3 per month	1st, 11th, 21st
Ceilings (petrol)	2.000	strict, from 28 Sep
Ceilings (diesel)	2.200 / 2.261 / 2.212	from 28 Sep / 6 Oct / 28 Oct

Table 2: Oil pass-through models: selection, out-of-sample fit and October price gaps

	Order	N	Placebo backtest 14–27 Sep (c/l)			Gap, 28 Sep–31 Oct (c/l)	
	(BIC)		RMSE	RMSE RW	Bias	Mean	31 Oct
Petrol	(2,8)	261	0.45	3.68	+0.01	−17.4	−22.4
Diesel	(1,8)	261	0.85	5.07	-0.17	−21.1	−20.5

Dependent variable: log national mean net-of-tax price (self-service, roadside); regressors: own lags and contemporaneous plus lagged log Brent in €. Estimation 2 Jan–27 Sep 2026 (261 observations after lags); BIC over $p \in [0, 21]$, $q \in [7, 21]$. Backtest: refit to 13 Sep, recursive projection over 14–27 Sep, compared with a no-change random walk. Gap: actual minus counterfactual gross price, days available (32 of 34; no extracts 16–17 Oct).

The results are in table 3. Observed reference-day petrol prices fell 2.3% month-on-month where the no-cap path implies a rise of 5.3%; diesel rose 1.6% where the no-cap path implies 10.7%. Weighted, the cap's estimated contribution to October headline NIC inflation is −0.38 percentage points (petrol −0.17, diesel −0.21). The assumption sensitivities are an order of magnitude smaller than the effect itself: 45/55 and 55/45 petrol–diesel splits give −0.381 and −0.374; the regularity-filtered sample gives −0.411; and across the specification range of section 5 the estimate lies between −0.38 and −0.45, with −0.30 under the inadmissible autoregression-only benchmark. The dominant uncertainty is the no-cap scenario, not the aggregation.

Two measurement notes complete the calculation. First, the communication-time reconstruction of section 3 matters directly for reference-day statistics: on 1 October the extract-time mean

Table 3: Reference-day inflation: actual versus no-cap, and the estimated contribution to headline NIC inflation

	Sep–Oct change (%)			Contribution
	Actual	No cap	Difference	(pp of NIC)
Petrol	−2.33	+5.28	−7.6	−0.17
Diesel	+1.58	+10.68	−9.1	−0.21
Motor fuels (combined)	−0.38	+7.98	−8.4	−0.38
Weight split 45/55 (petrol/diesel)				−0.38
Weight split 55/45 (petrol/diesel)				−0.37
Regularity-filtered stations (50/50)				−0.41

Month-on-month changes over the 1st/11th/21st reference days, national mean gross prices, equal station weights. Contributions use the NIC 2026 weight of ECOICOP 07.2.2 (4.5219%, verified from ISTAT's published weighting structure) split evenly between petrol and diesel. September no-cap prices equal observed prices by construction.

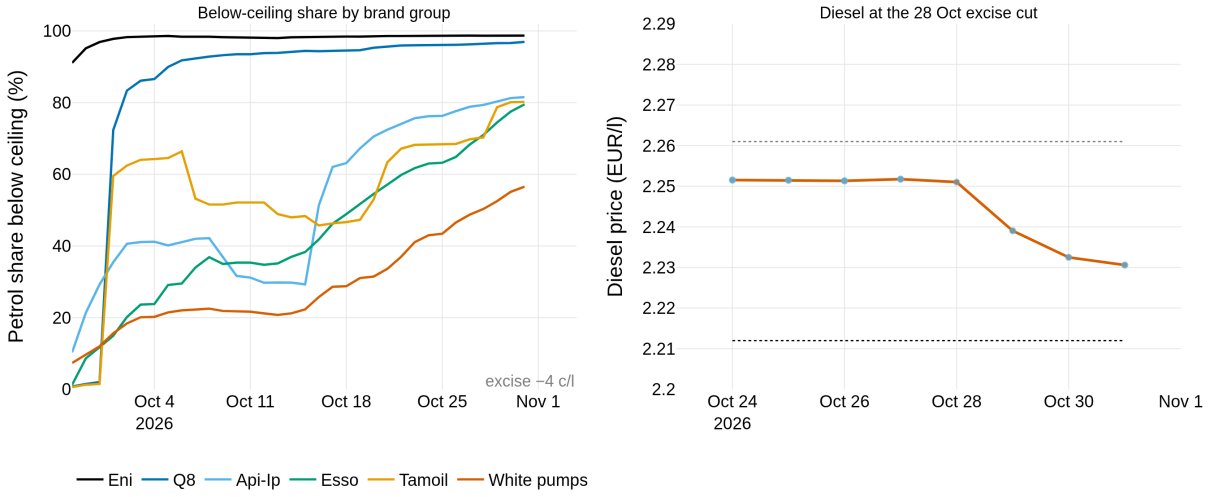


Figure 4: Propagation. Left: share of petrol observations below the ceiling by brand group, 28 September–31 October. Right: the 28 October diesel episode—mean diesel price day by day (markers shaded by the below-ceiling share), with the old ceiling (2.261, grey) and the mechanically adjusted ceiling (2.212, black).

overstated the effective price in force by 4.2–4.9 c/l. This is a correction to snapshot-based measurement of our reconstructed estimator; we have no basis, and no need, to attribute error to the published index, whose acquisition procedures may differ. Second, fuel-price components are, in the authors' institutional experience, complete by the preliminary release, so we do not treat the final release as a condition for these estimates; this premise is stated as such.

7 How the response propagated

The aggregate price decline was produced by a rapid, ordered, geographically broad repricing, and the station panel shows its anatomy. Figure 4a traces the below-ceiling share of petrol observations by brand group: the initiator's network prices at the ceiling from the first days; Q8 and Api-Ip converge within a week; Esso and Tamoil slightly later; white-pump independents partially. Timing correlates with local market structure exactly as conditional-association analysis would have it: in duration models (table 4), a station ten cents closer to the ceiling in the pre-cap week adopts sooner (log-hazard −0.44 per 10 c/l in the universe, −0.61 to −0.66 once the leader brands are excluded or among independents), adoption is slower the farther the station is from

its nearest competitor (−0.10 log-hazard per kilometre) and from the nearest day-zero adopter (−0.03 to −0.06 per kilometre), and larger chains adopt sooner (the global proportional-hazards restriction is rejected, so these are time-averaged associations rather than constant hazards; we read them as descriptive propagation gradients, not causal effects). A discrete-time logit hazard with day fixed effects, which requires no proportional-hazards assumption, reproduces every gradient (table 12): per ten cents of pre-cap gap the daily odds of adopting fall by a factor of about 0.41 in the universe and 0.21 among white pumps, with the distance and chain gradients likewise intact, and its baseline hazard spikes exactly at the 1 October repricing wave and the 6 October excise step. The station-fixed-effects event study (table 5) shows flat pre-cap coefficients (−0.002 to −0.004, an order of magnitude smaller than the day-zero jump of +0.19) and a plateau after four to five days.

The ceiling became the market's focal price—not the announced nominal level, but the level one tenth of a cent below the operational cutoff. After the cap, 17.9% of all petrol observations sit at exactly 1.999 €/l; the share of diesel observations ending in .199 rises from 2.0% before the cap to 11.2% after it (table 9). When the 6 October excise step moved the diesel ceiling to 2.261, the mass re-formed at 2.259—eight and a half per cent of observations—rather than at the announced-style 2.19. This is focal-point pricing of the kind documented for statutory ceilings (Gatsios et al., 2026; Zhang et al., 2020), here generated by a voluntary commitment and matched by rivals.

Dispersion effects are best stated with numbers, because different summaries point in different directions. The within-province tenth-to-ninetieth percentile spread rose from about 8.1 to 14.0 c/l (petrol) and 9.5 to 15.9 c/l (diesel) between the full pre-cap month and the cap window—an increase of roughly seventy per cent. Standard deviations rose for petrol (4.8 to 7.1 c/l within province–day) and for diesel relative to the tighter pre-cap week, but fell for diesel relative to the full September baseline, because the pre-cap diesel distribution was itself unusually dispersed after the two mid-September excise steps. The economically meaningful statistic is the gap between adopters and non-adopters: within province–day, stations below the ceiling priced on average 11.5 (petrol) and 11.7 c/l (diesel) below stations above it, and the non-adopter tail drifted upward with oil exactly as the counterfactual of section 5 predicts.

The window's final days deliver a small natural experiment in ceiling-constrained tax incidence. When the 28 October excise cut lowered the mechanical ceiling from 2.261 to 2.212, the mean diesel price did not move on the day (2.258 on both 27 and 28 October); by 31 October it had fallen only 1.96 c/l of the 4.88 c/l mechanical adjustment —a pass-through of about forty per cent in three days. The below-ceiling share collapsed from 65.8% to 24.0% on the day of the cut and recovered to 50.2% by 31 October; not a single observation priced at 2.211, the new bunching point, in the final three days. With the commitment on the cap level running out, the ceiling stopped functioning as a focal price and the network absorbed part of the tax cut instead of passing it through—the incomplete, asymmetric pass-through familiar from the rockets-and-feathers literature (Borenstein & Shepard, 2002; Tappata, 2009), here observed at a ceiling. Brand groups, however, read the same episode very differently (table 10). Eni's network had been pricing at the announced nominal level all along and ended October at 2.190 €/l on average—compliant under any ceiling in force, with no repricing required of it. The gradient appears entirely among the followers: Q8 cut 3.8 c/l and reached 78% compliance under the new ceiling, Api-Ip 2.7 c/l and 60%, Esso and Tamoil between 2.4 and 3.4 c/l with 22–32%, and unbranded independents cut least (1.8 c/l, 15% compliant). The leader treated the commitment level as binding to the end; the followers' partial pass-through is what the aggregate incompleteness is made of.

Table 4: Cox duration-model estimates by sample (coefficients; hazard ratios = $e^{\hat{\beta}}$)

	(1) Universe	(2) Excl. Agip Eni + Q8	(3) Independents
Nearest station (km)	-0.1001***	-0.0874***	-0.1340***
Nearest day-0 adopter (km)	-0.0317***	-0.0531***	-0.0642***
Population (1,000 per km^2 cell)	0.008**	0.007*	-0.001
Pre-cap gap (10 c/l)	-0.439***	-0.664***	-0.613***
Chain size (10 stations)	0.006***	0.009***	0.055***
Diesel	0.0744***	0.1006**	0.0561
N	39,423	26,722	5,192
Events	32,971	20,438	3,027

Gestore-clustered robust standard errors. All models include a diesel dummy. The diesel cap threshold is adjusted for the excise change of 6 October 2026 (see section 3).

8 Conclusion

A voluntary price ceiling, announced by the market leader during an oil surge and matched by most of the network within two weeks, held Italian pump prices far below the level implied by the pre-cap relationship between oil and pump prices: on our baseline estimates, 17 to 21 cents per litre below over October, with an estimated contribution to October headline consumer-price inflation of about -0.4 percentage points under transparent aggregation assumptions. The station-level panel shows how the effect was produced: a leader-led, competition-gradient repricing that bunched prices one tenth of a cent below the ceiling and left a non-adopter tail drifting upward with oil, widening the price distribution while lowering its mean.

The conclusions are conditional, and their limits are the paper's caveats. The counterfactual assumes pre-cap pass-through would have persisted; we validate it where possible—placebo and expanding-origin forecasts that beat random-walk benchmarks, oil blocks that cannot be excluded, specification ranges that bound the answer—but a model-based no-cap path is not an experiment. The window ends on 31 October with the commitment's hard horizon; the gradual pass-through of the final diesel excise cut suggests the ceiling's force was already fading. And the micro gradients we document are conditional associations. What the episode establishes is measurement: the size of a voluntary cap's price and inflation effects can be bounded credibly from a complete price-communication registry, and the details of that registry—communication times, three-decimal quoting, tax-translated ceilings—matter for every number in it.

A Additional tables

Table 5: Station-fixed-effects event study: adoption below the ceiling (leads and lags)

Event time	$\hat{\beta}_k$	Std. error
Day -7	-0.002	0.001
Day -6	-0.002	0.001
Day -5	-0.003*	0.001
Day -4	-0.003*	0.001
Day -3	-0.004**	0.001
Day -2	-0.004**	0.001
Day +0	+0.193***	0.037
Day +1	+0.243***	0.037
Day +2	+0.277***	0.037
Day +3	+0.451	0.046
Day +4	+0.492	0.044
Day +5	+0.507	0.042
Day +6	+0.509	0.042
Day +7	+0.527	0.040

Dependent variable: indicator for pricing below the ceiling; station–fuel fixed effects; gestore-clustered errors; day −1 (27 September) omitted.

B Levels relationship: bounds tests

The main text reports long-run elasticities as a range and leans on forecast validation rather than on a point estimate of long-run pass-through. This appendix formalises that choice. We re-estimate the selected models in error-correction form and apply the Pesaran–Shin–Smith bounds procedure (Pesaran et al., 2001) to the null of no level relationship between the log net-of-tax price and log Brent in euro (case 3: unrestricted intercept, no trend), on the same 2 January–27 September window as the counterfactual.

Table 11 reports the results. For petrol the bounds F-statistic is 1.40 and for diesel 0.92, both far below their $I(0)$ lower critical bounds of about 4.9; the t-statistics (-1.63 and -0.60) are likewise indistinguishable from the no-level- relationship region. The implied error-correction coefficients are small and insignificant (-0.013 and -0.005 per day). Over this window, in short, net pump prices and Brent share no detectable levels relationship: pass-through works through short-run dynamics, and any long-run elasticity computed from the levels representation of these models—including the diesel main specification's small-denominator value of 0.10—is not an identified object. This is consistent with the near-unit-root persistence documented in section 5 and is why the counterfactual's credibility rests on its validated multi-step forecasts rather than on a cointegration claim. A longer history including pre-2026 data might identify a levels relationship; within 2026, the oil shock, the spring excise cycle and the cap episode are too entangled to deliver one.

Table 6: Specification grid and scenarios: October mean gaps and implied NIC effects

		Order (BIC)	Backtest RMSE (c/l)	Mean gap (c/l)	NIC effect (pp)
Petrol	No oil-lag floor	(1,2)	1.60	−17.5	−0.171
Petrol	Min 2 oil lags	(1,2)	1.60	−17.5	−0.171
Petrol	Min 5 oil lags	(1,5)	1.53	−17.5	−0.169
Petrol	Min 7 oil lags (main)	(2,8)	0.45	−17.4	−0.172
Petrol	Distributed lag only	(0,4)	4.88	−11.4	−0.091
Petrol	Autoregression only (no oil)	(1,.)	3.00	−14.3	−0.145
Diesel	No oil-lag floor	(3,3)	2.30	−22.7	−0.212
Diesel	Min 2 oil lags	(3,3)	2.30	−22.7	−0.212
Diesel	Min 5 oil lags	(3,5)	2.39	−22.2	−0.207
Diesel	Min 7 oil lags (main)	(1,8)	0.85	−21.1	−0.206
Diesel	Distributed lag only	(0,4)	3.06	−26.1	−0.219
Diesel	Autoregression only (no oil)	(3,.)	4.31	−15.3	−0.155
Petrol	fit from Jan (baseline)	(2,8)	–	−17.4	−0.172
Diesel	fit from Jan (baseline)	(1,8)	–	−21.1	−0.206
Petrol	fit from Jul (v1 window)	(4,7)	–	−16.3	−0.161
Diesel	fit from Jul (v1 window)	(1,7)	–	−24.5	−0.236
Petrol	fit to 24 Sep (announcement)	(2,8)	–	−16.3	−0.163
Diesel	fit to 24 Sep (announcement)	(1,8)	–	−23.1	−0.228
Petrol	balanced panel series	(2,8)	–	−17.6	−0.173
Diesel	balanced panel series	(1,8)	–	−21.2	−0.206
Petrol	regularity sample series	(2,8)	–	−18.8	−0.189
Diesel	regularity sample series	(1,8)	–	−23.6	−0.231
Petrol	Brent +5% post-cap	(2,8)	–	−19.7	−0.194
Diesel	Brent +5% post-cap	(1,8)	–	−24.2	−0.235
Petrol	Brent -5% post-cap	(2,8)	–	−15.1	−0.150
Diesel	Brent -5% post-cap	(1,8)	–	−18.0	−0.175

Upper block: oil-lag floors and distributed-lag / autoregression-only extremes (all re-selected by BIC on the extended window, max lag 21). Lower block: fit window, sample and Brent-path scenarios. The autoregression-only row is a mechanical persistence benchmark, not an admissible no-oil counterfactual. NIC effects per fuel use the 50/50 split.

Table 7: Expanding-origin backtests (all evaluation windows end before 28 September)

	Origin	Horizon (days)	RMSE (c/l)	RMSE RW (c/l)	Bias (c/l)
Petrol	31 May	14	2.48	2.54	+2.35
Petrol	31 May	28	4.24	6.38	+3.82
Petrol	30 Jun	14	2.96	1.21	+2.52
Petrol	30 Jun	28	5.13	5.15	+4.56
Petrol	31 Jul	14	1.33	0.67	+1.19
Petrol	31 Jul	28	1.96	0.86	+1.75
Petrol	31 Aug	14	0.66	4.58	+0.63
Petrol	13 Sep	14	0.45	3.68	+0.01
Diesel	31 May	14	2.62	3.93	-2.19
Diesel	31 May	28	2.94	8.51	+0.09
Diesel	30 Jun	14	3.67	2.37	+3.10
Diesel	30 Jun	28	7.14	11.29	+6.17
Diesel	31 Jul	14	1.68	0.76	+1.54
Diesel	31 Jul	28	1.95	2.07	+1.80
Diesel	31 Aug	14	4.74	4.48	-3.79
Diesel	13 Sep	14	0.85	5.07	-0.17

Models re-selected by BIC on each subsample; recursive projection over the stated horizon; comparison against a no-change random walk.

Table 8: Exact prices at the ceiling and in the bunching bin below it

Fuel	Ceiling	Price	Obs.	Share
Petrol	2.0	1.998	2 940	0.45%
Petrol	2.0	1.999	108 281	16.69%
Petrol	2.0	2.0	143	0.02%
Petrol	2.0	2.001	33	0.01%
Diesel	2.2	2.198	173	0.11%
Diesel	2.2	2.199	10 310	6.35%
Diesel	2.2	2.2	14	0.01%
Diesel	2.2	2.201	0	0.00%
Diesel	2.212	2.21	107	0.13%
Diesel	2.212	2.211	0	0.00%
Diesel	2.212	2.212	4	0.01%
Diesel	2.212	2.213	0	0.00%
Diesel	2.261	2.259	34 332	8.46%
Diesel	2.261	2.26	665	0.16%
Diesel	2.261	2.261	229	0.06%
Diesel	2.261	2.262	44	0.01%

Shares within each fuel–ceiling segment; compliance is strict, so the ceiling price itself is not compliant. The bunching bin (one tenth of a cent below the ceiling: 1.999, 2.199, 2.260, 2.211) concentrates up to 17.9% of petrol observations.

Table 9: Most frequent price endings before and after the cap (share of observations)

Window	Fuel	Top endings (cent-decimal: share)		
Pre-cap (1–27 Sep)	Petrol	.10: 3.9%	.15: 3.7%	.16: 3.5%
Pre-cap (1–27 Sep)	Diesel	.20: 3.7%	.17: 2.5%	.35: 2.5%
Post-cap (28 Sep–31 Oct)	Petrol	.99: 17.5%	.00: 11.0%	.10: 3.0%
Post-cap (28 Sep–31 Oct)	Diesel	.19: 11.1%	.25: 7.6%	.30: 4.5%

Table 10: Brand groups at the 28 October ceiling step (diesel)

		27 Sep → 31 Oct		Below 2.212 ceiling	
Group	Stations	Δ price (c/l)	Level 31 Oct	28 Oct	31 Oct
Eni	3 741	-0.0	2.190	99%	99%
Q8	2 584	-3.8	2.216	2%	78%
Api-Ip	3 385	-2.7	2.229	5%	60%
Esso	1 890	-2.4	2.250	8%	32%
Tamoil	1 374	-3.4	2.240	3%	22%
Other brands	4 032	-1.6	2.271	9%	20%
White pumps	2 584	-1.8	2.286	10%	15%

Mean gross diesel price change from 27 October (last day of the 2.261 ceiling) to 31 October, and the share of observations strictly below the mechanically adjusted 2.212 ceiling on 28 and 31 October. Eni's network was already at the announced nominal level (2.19), hence compliant under any ceiling in force.

Table 11: Pesaran–Shin–Smith bounds tests for a level relationship (net price, Brent), case 3, estimation window 2 Jan–27 Sep 2026

	Order	F statistic	5% bounds $[I(0), I(1)]$	t statistic	ECT (per day)	Implied half-life
Petrol	(2,8)	1.40	[4.92, 5.74]	-1.63	-0.0134	52 days
Diesel	(1,8)	0.92	[4.92, 5.74]	-0.60	-0.0047	148 days

F and t test the null of no level relationship between the log net-of-tax price and log Brent in euro; critical bounds from Pesaran et al. (2001). ECT: coefficient on the lagged level in the unrestricted error-correction representation. Both statistics lie far below their $I(0)$ lower bounds: over this window the relationship is a short-run one, and levels long-run elasticities are not identified — which is why the paper reports elasticity ranges and rests the counterfactual on forecast validation.

Table 12: Discrete-time logit hazard of first pricing below the ceiling

	(1) Universe	(2) Excl. Eni + Q8	(3) White pumps
Pre-cap gap (10 c/l)	-0.888***	-1.143***	-1.551***
Nearest station (km)	-0.072***	-0.080***	-0.159***
Nearest day-0 adopter (km)	-0.040***	-0.053***	-0.051***
Population (1,000 per km^2 cell)	-0.000	+0.001	+0.006
Chain size (10 stations)	+0.004***	+0.003	+0.057
Diesel	+0.086**	+0.074*	-0.076
Day fixed effects	Yes	Yes	Yes
Brand fixed effects	Yes	Yes	– (single brand)
Region fixed effects	Yes	Yes	Yes

One row per station–fuel unit per day at risk (463,594 unit-days; 32,974 adoptions); the event is the first day below the ceiling. Log-odds; gestore-clustered standard errors suppressed for readability (all starred coefficients significant at 0.1% or better; population insignificant everywhere). Day fixed effects absorb the baseline hazard, which makes no proportional-hazards assumption necessary; the estimated baseline spikes on the 1 October repricing wave and the 6 October excise step.

Table 13: Adoption probability (LPM) by sample — core covariates

	(1) Universe	(2) Excl. Agip Eni + Q8	(3) Independents
Intercept	1.2124***	0.8733***	0.8622***
Nearest station (km)	-0.0227***	-0.0246***	-0.0283***
Nearest day-0 adopter (km)	-0.0117***	-0.0164***	-0.0171***
Population (1,000 per km^2 cell)	-0.001	-0.001	0.002
Pre-cap gap (10 c/l)	-0.109***	-0.138***	-0.105***
Chain size (10 stations)	0.000	0.001*	0.003
Diesel	-0.0241***	-0.0402***	-0.0464**
Brand & region FE	Yes	Yes	Region only
N	39272	26622	5164
Adj. R^2	0.2446	0.2143	0.1913

Dependent variable: adopted (=1 if the station–fuel unit priced at or below the cap on any day 28 Sep – 31 Oct 2026; the diesel cap is tax-adjusted from 6 Oct). OLS with gestore-clustered standard errors. Brand reference: Agip Eni in columns (1), Ala in column (2); region reference: Abruzzo. Full brand and region coefficients in tables 14 and 15. In column (3) the sample is single-brand (independents), so brand FE are undefined.

Table 14: Adoption probability (LPM), universe sample — brand and region fixed effects

	Coefficient		Coefficient
Brand: Ala	-0.4095***	Brand: Sia fuel	-0.0490*
Brand: Api-Ip	-0.1094***	Brand: Sommese Petroli	-0.3684***
Brand: Aquila	0.0010	Brand: Tamoil	-0.1355***
Brand: Beyfin	-0.1223*	Brand: Toil	-0.2667**
Brand: COIL	-0.1058*	Brand: Vega	-0.0155*
Brand: CONAD	-0.0508**	Region: (no region)	-0.1212
Brand: Costantin	-0.0416	Region: Basilicata	-0.0545
Brand: Ego	-0.1811**	Region: Calabria	-0.0261
Brand: Energas	-0.0286	Region: Campania	-0.0425*
Brand: Enerpetroli	-0.1004***	Region: Emilia-Romagna	0.0055
Brand: Esso	-0.1490***	Region: Friuli-Venezia Giulia	-0.0655*
Brand: Europam	-0.0924	Region: Lazio	0.0138
Brand: Giap	-0.2239***	Region: Liguria	0.0252
Brand: Itala Petroli	-0.2085***	Region: Lombardia	0.0389*
Brand: KEROPETROL	-0.4234***	Region: Marche	0.0066
Brand: Loro	-0.5435***	Region: Molise	-0.0846**
Brand: OTHER	-0.2524***	Region: Piemonte	0.0485*
Brand: Petrol Gamma	-0.3936***	Region: Puglia	-0.0124
Brand: Independents	-0.3874***	Region: Sardegna	0.0546**
Brand: Q8	-0.0179	Region: Sicilia	-0.0024
Brand: Retitalia	-0.2677***	Region: Toscana	-0.0103
Brand: San Marco Petroli	-0.6682***	Region: Umbria	0.0259
Brand: Shell	-0.1229***	Region: Veneto	0.0178

Continuation of table 13 column (1). Brand reference: Agip Eni; region reference: Abruzzo. Brands with fewer than 100 observations are grouped into OTHER.

Table 15: Adoption probability (LPM), excluding Agip Eni and Q8 — brand and region fixed effects

	Coefficient		Coefficient
Brand: Api-Ip	0.2885***	Brand: Sommese Petroli	0.0206
Brand: Aquila	0.4172***	Brand: Tamoil	0.2496***
Brand: Beyfin	0.2902***	Brand: Toil	0.1472
Brand: COIL	0.2826***	Brand: Vega	0.3729***
Brand: CONAD	0.3442***	Region: (no region)	-0.1113
Brand: Costantin	0.3701***	Region: Basilicata	-0.0763
Brand: Ego	0.2201***	Region: Calabria	-0.0418
Brand: Energas	0.3717***	Region: Campania	-0.0469
Brand: Enerpetroli	0.3028***	Region: Emilia-Romagna	0.0115
Brand: Esso	0.2563***	Region: Friuli-Venezia Giulia	-0.1547***
Brand: Europam	0.2985***	Region: Lazio	0.0128
Brand: Giap	0.1977***	Region: Liguria	0.0391
Brand: Itala Petroli	0.1979***	Region: Lombardia	0.0568*
Brand: KEROPETROL	-0.0243	Region: Marche	0.0065
Brand: Loro	-0.1362**	Region: Molise	-0.1240**
Brand: OTHER	0.1564***	Region: Piemonte	0.0689*
Brand: Petrol Gamma	0.0212	Region: Puglia	-0.0207
Brand: Independents	0.0203	Region: Sardegna	0.0758*
Brand: Retitalia	0.1362**	Region: Sicilia	-0.0161
Brand: San Marco Petroli	-0.2370***	Region: Toscana	-0.0183
Brand: Shell	0.2681***	Region: Umbria	0.0408
Brand: Sia fuel	0.3622***	Region: Veneto	0.0164

Continues table 13 column (2). Brand reference: Ala; region reference: Abruzzo. Brands with fewer than 100 observations are grouped into OTHER.

C Additional estimates

D Provincial maps

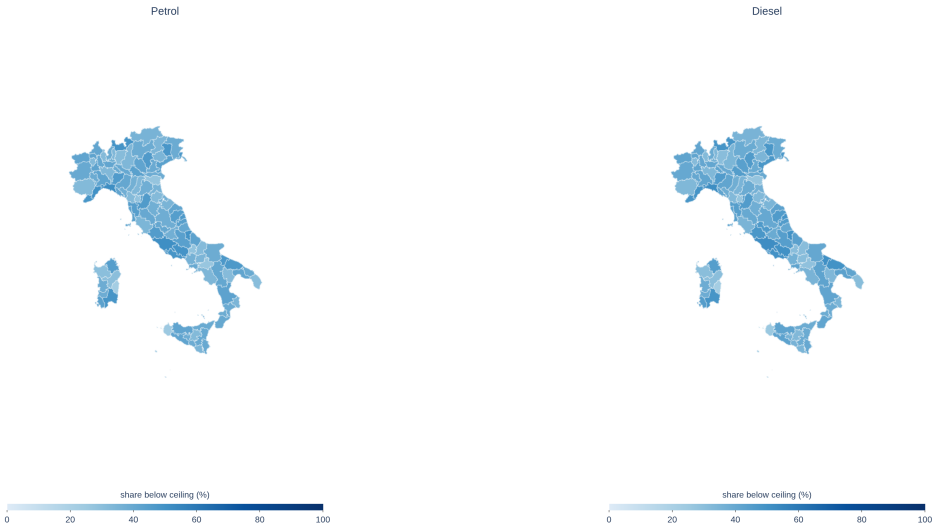


Figure 5: Share of station–fuel units priced below the ceiling at least once in the first seven days (28 September–4 October), by province: petrol (left) and diesel (right); common 0–100% scale. Early diffusion is geographically uneven but fuel-symmetric (province medians 39–40%).

E Data dictionary

Table 16: Variable names used in the replication materials and their paper terms.

Code name	Paper term
`dt_comu` / `dtComu`	communication timestamp
`prezzo`	pump (gross) price, EUR/l to three decimals
`eff_mean_gross` / `eff_mean_net`	national mean effective gross / net-of-tax price
`cap_threshold`	date-specific operational ceiling
`at_cap` / `compliance_at_cap`	indicator: effective price strictly below the ceiling
`nn_dist_km`	distance to nearest priced competitor (km)
`cap_dist_km_d0`	distance to nearest below-ceiling station on 28 September (km)
`pre_gap_vs_cap`	pre-cap-week mean price minus nominal ceiling (EUR/l)
`pop_density_cell`	population of the station's 1-km grid cell
`n_stations_gestore`	number of stations under the same operator
`bandiera`	brand banner ("Agip Eni" = the initiator; "Pompe Bianche" = white pumps)

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